

## BPHO ROUND 0 · QUESTION PAPER

# Drill bank section 5

25 questions · 60 minutes · single answer, 5 options · **no calculator** · no negative marking

*Original questions written for this course, in the Round 0 format. Not BPhO material.*

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**Before you start**

- Answer every question. There is no negative marking, so a blank is a wasted mark.
- No calculator. Every answer here is reachable by hand, and the numbers are chosen to divide cleanly.
- Work on paper, then mark yourself against the markscheme PDF.
- The 2025 UK qualifying line was 11 out of 25, so about 44% is the target to beat.

**Q1** · Dimensional analysis: which combination has the dimensions of pressure A

Which one of the following has the same dimensions as pressure (force divided by area)?

**A** energy / volume

**B** power / area

**C** force / length

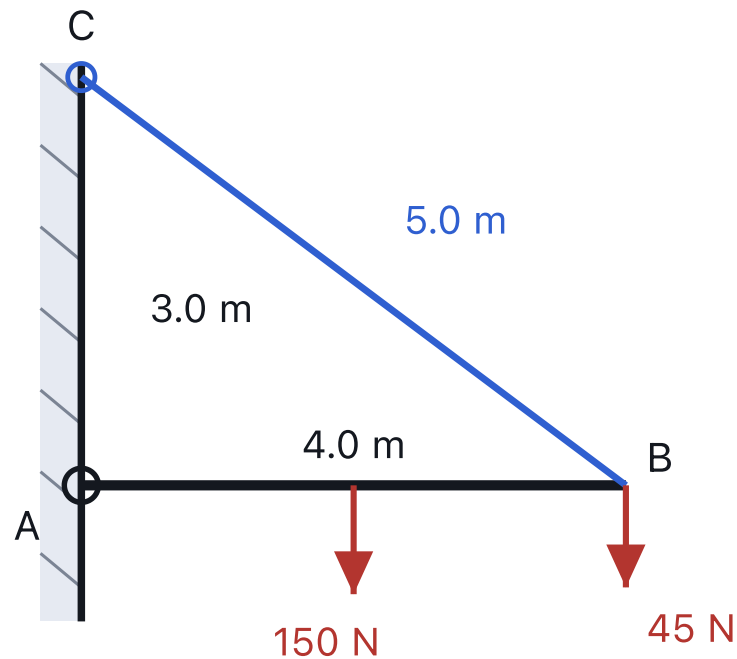
**D** momentum / volume

**E** mass / (area x time)

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**Q2** · A hinged rod held by a cable: the tension from moments C

A uniform-free horizontal rod **AB** is 4.0 m long and is hinged to a wall at **A**. A cable joins **B** to a point **C** that is 3.0 m vertically above **A**; the cable is therefore 5.0 m long. A weight of 150 N acts at the middle of the rod and a load of 45 N hangs from **B**.

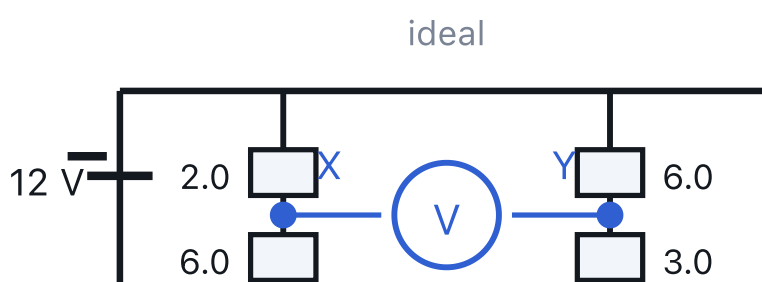


What is the tension in the cable?

- |                |                |
|----------------|----------------|
| <b>A</b> 150 N | <b>B</b> 250 N |
| <b>C</b> 200 N | <b>D</b> 120 N |
| <b>E</b> 160 N |                |
-

**Q3** · Two potential dividers: the voltmeter reading between their midpoints H

A 12 V supply feeds two branches in parallel. The left branch is 2.0 ohm above its midpoint X and 6.0 ohm below it; the right branch is 6.0 ohm above its midpoint Y and 3.0 ohm below it. An ideal voltmeter bridges X and Y.



What does the voltmeter read?

- |                |                |
|----------------|----------------|
| <b>A</b> 3.0 V | <b>B</b> 8.0 V |
| <b>C</b> 9.0 V | <b>D</b> 1.5 V |
| <b>E</b> 5.0 V |                |

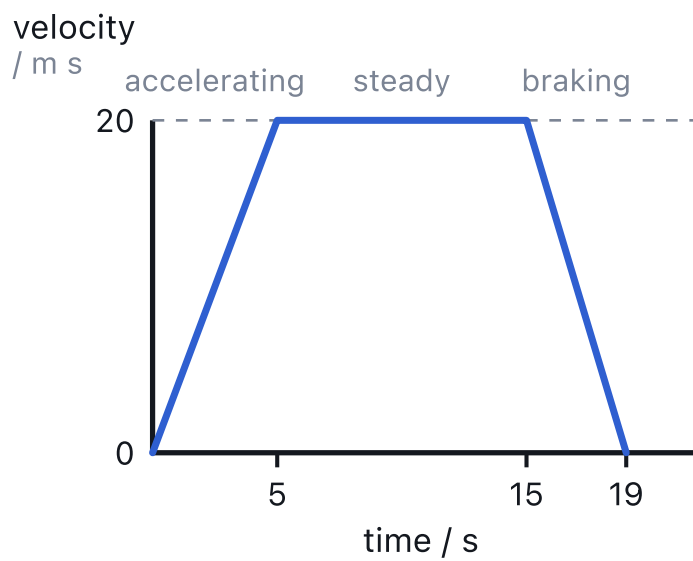
**Q4** · Order of magnitude: the number of seconds in one year A

To the nearest power of ten, how many seconds are there in one year?

- |                 |                 |
|-----------------|-----------------|
| <b>A</b> $10^5$ | <b>B</b> $10^7$ |
| <b>C</b> $10^6$ | <b>D</b> $10^8$ |
| <b>E</b> $10^9$ |                 |

**Q5** · A velocity-time graph: the total distance from the area **B**

A vehicle moves along a straight line. Its velocity-time graph is shown.

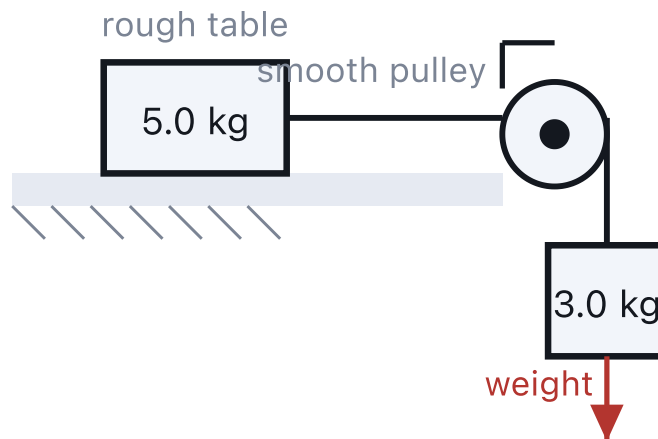


What total distance does it travel between  $t = 0$  and  $t = 19$  s?

- |                |                |
|----------------|----------------|
| <b>A</b> 250 m | <b>B</b> 300 m |
| <b>C</b> 200 m | <b>D</b> 290 m |
| <b>E</b> 340 m |                |
-

**Q6** · A block on a table linked over a pulley: the normal reaction C

A 5.0 kg block rests on a rough horizontal table and is joined by a light string over a smooth pulley to a hanging 3.0 kg block.

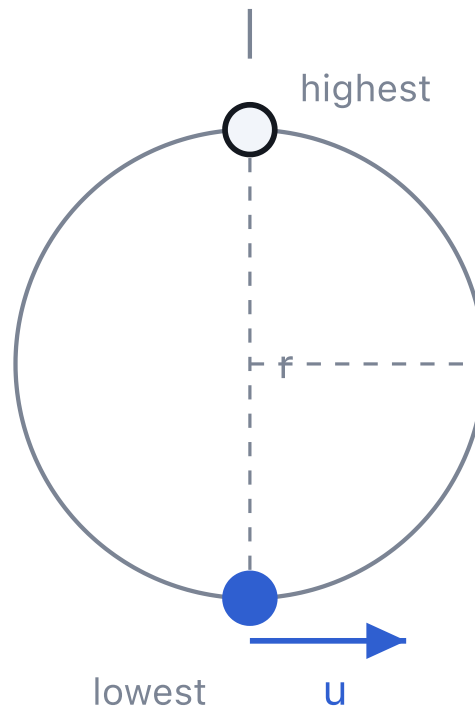


What is the normal reaction of the table on the 5.0 kg block? (Use  $g = 9.8 \text{ m/s}^2$ )

- |               |               |
|---------------|---------------|
| <b>A</b> 20 N | <b>B</b> 78 N |
| <b>C</b> 29 N | <b>D</b> 49 N |
| <b>E</b> 50 N |               |
-

**Q7** · A vertical circle: the minimum speed at the bottom for the string to stay taut D

A particle of mass  $m$  is tied to a light string of length  $r$  and whirled in a vertical circle. Its speed at the lowest point is  $u$ . The string must remain taut all the way round, including at the highest point, which is two radii above the lowest.



What is the minimum value of  $u$  for this to happen?

**A**  $\sqrt{(5 \text{ g } r)}$

**B**  $\sqrt{(2 \text{ g } r)}$

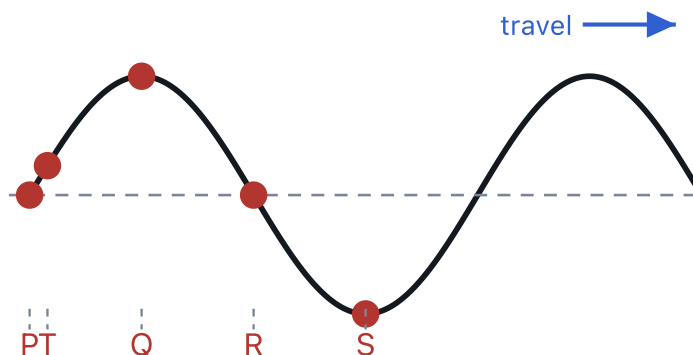
**C**  $\sqrt{(3 \text{ g } r)}$

**D**  $\sqrt{(g \text{ r})}$

**E**  $\sqrt{(4 \text{ g } r)}$

**Q8** · A wave snapshot: the transverse velocity from the local slope **F**

The diagram is a snapshot of a transverse wave of amplitude 2.0 cm travelling to the right at 1.5 m/s. At the point labelled Q the string makes an angle whose slope is -0.40.

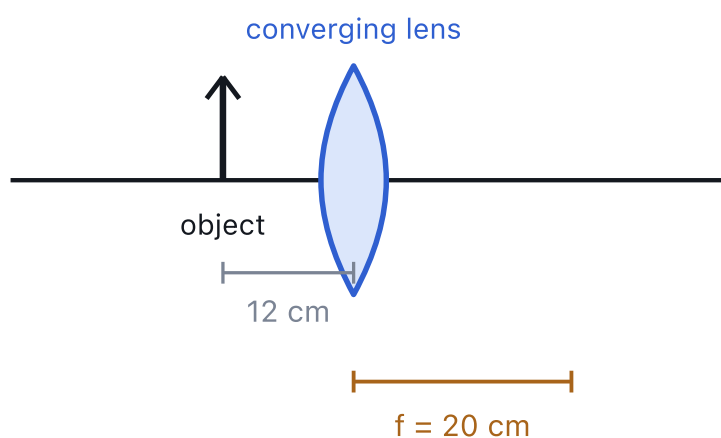


What is the transverse velocity of the point Q?

- |                            |                          |
|----------------------------|--------------------------|
| <b>A</b> 0.60 m/s downward | <b>B</b> 0.40 m/s upward |
| <b>C</b> 0.60 m/s upward   | <b>D</b> 0.90 m/s upward |
| <b>E</b> 0 m/s             |                          |

**Q9** · A converging lens: an object inside the focal length gives a virtual image **G**

An object is placed 12 cm to the left of a converging lens of focal length 20 cm.

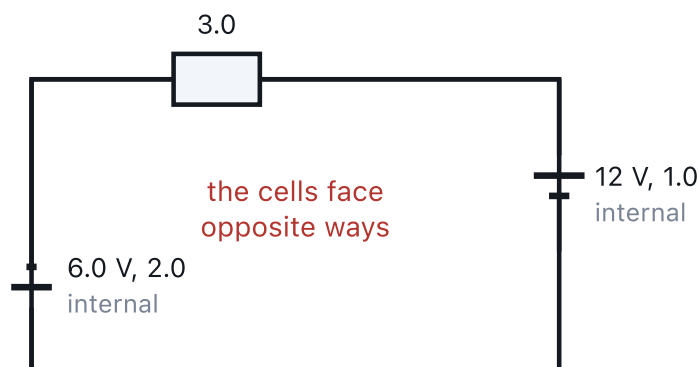


What is the linear magnification produced by the lens?

- |               |               |
|---------------|---------------|
| <b>A</b> 0.40 | <b>B</b> 1.7  |
| <b>C</b> 2.0  | <b>D</b> 0.60 |
| <b>E</b> 2.5  |               |

**Q10** · Two opposing cells: the current in the single loop H

A single loop contains a 3.0 ohm resistor, a 12 V cell of internal resistance 1.0 ohm with its positive terminal uppermost, and a 6.0 V cell of internal resistance 2.0 ohm with its positive terminal lowermost, so the two cells oppose each other.



What is the current in the circuit?

- |                                                                   |                                              |
|-------------------------------------------------------------------|----------------------------------------------|
| <p><b>A</b> 2.0 A</p> <p><b>C</b> 3.0 A</p> <p><b>E</b> 6.0 A</p> | <p><b>B</b> 1.0 A</p> <p><b>D</b> 0.50 A</p> |
|-------------------------------------------------------------------|----------------------------------------------|
- 

**Q11** · Radioactive decay: the fraction remaining after a number of half-lives K

A radioactive sample has a half-life of 8.0 days. What fraction of the original nuclei remains after 24 days?

- |                                                              |                                         |
|--------------------------------------------------------------|-----------------------------------------|
| <p><b>A</b> 1/4</p> <p><b>C</b> 1/16</p> <p><b>E</b> 1/2</p> | <p><b>B</b> 1/8</p> <p><b>D</b> 3/8</p> |
|--------------------------------------------------------------|-----------------------------------------|
- 

**Q12** · The momentum of a photon of given wavelength L

What is the momentum of a photon of wavelength 600 nm? (Use  $h = 6.6 \times 10^{-34} \text{ J s}$ .)

- |                                                                                                                                                                                                                |                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>A</b> <math>3.3 \times 10^{-19} \text{ kg m s}^{-1}</math></p> <p><b>C</b> <math>1.1 \times 10^{-19} \text{ kg m s}^{-1}</math></p> <p><b>E</b> <math>6.6 \times 10^{-34} \text{ kg m s}^{-1}</math></p> | <p><b>B</b> <math>5.5 \times 10^{-28} \text{ kg m s}^{-1}</math></p> <p><b>D</b> <math>1.1 \times 10^{-27} \text{ kg m s}^{-1}</math></p> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
-

**Q13** · Converting a speed from m/s to km/h **A**

A sprinter covers 100 m in 10 s. What is the average speed in km/h?

- A** 36 km/h
  - B** 10 km/h
  - C** 3.6 km/h
  - D** 360 km/h
  - E** 100 km/h
- 

**Q14** · A horizontal projectile: the range from the time of fall **B**

A ball is thrown horizontally at 12 m/s from a height of 20 m above level ground. (Use  $g = 10 \text{ m/s}^2$ .) How far from the point directly below the launch does it land?

- A** 12 m
  - B** 48 m
  - C** 24 m
  - D** 34 m
  - E** 17 m
- 

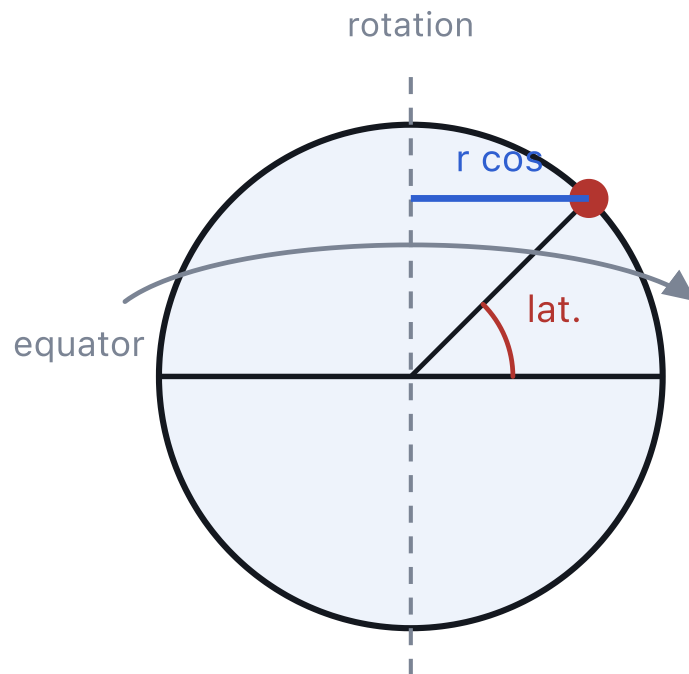
**Q15** · Two perpendicular forces: the equilibrating force **C**

Two forces, 3 N and 4 N, act at right angles on a body. What single force is needed to keep the body in equilibrium?

- A** 7 N
  - B** 1 N
  - C** 12 N
  - D** 3.5 N
  - E** 5 N
-

**Q16** · A rotating sphere: how latitude changes the centripetal acceleration D

A sphere of radius  $R$  rotates with angular speed  $\omega$  about a vertical axis. A point at latitude  $\lambda$  moves in a circle about that axis.

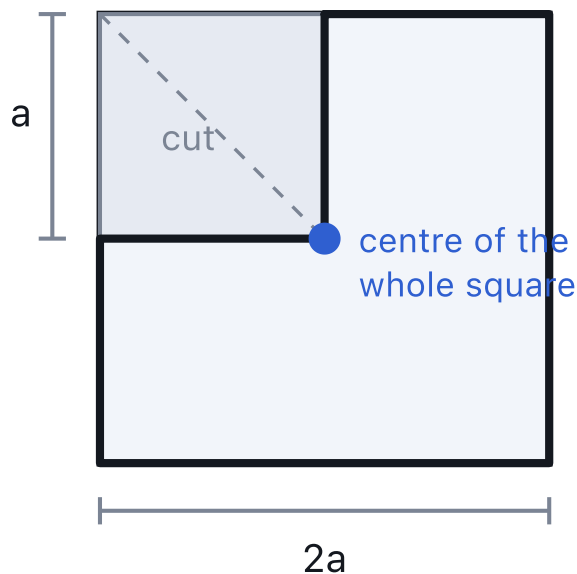


By what factor is the centripetal acceleration of that point less than the centripetal acceleration of a point on the equator?

- |                           |                           |
|---------------------------|---------------------------|
| <b>A</b> $\sin \lambda$   | <b>B</b> $\cos^2 \lambda$ |
| <b>C</b> $\sin^2 \lambda$ | <b>D</b> 1                |
| <b>E</b> $\cos \lambda$   |                           |

**Q17** · A square lamina with a corner removed: the shift of the centre of mass E

A uniform square lamina has side  $2a$ . A square of side  $a$  is removed from its upper-left corner.



By what distance is the centre of mass displaced horizontally from the centre of the original uncut square?

- |                 |                |
|-----------------|----------------|
| <b>A</b> $a/3$  | <b>B</b> $a/6$ |
| <b>C</b> $a/4$  | <b>D</b> $a/2$ |
| <b>E</b> $a/12$ |                |

**Q18** · Two coherent sources: the fringe spacing F

Two coherent sources  $1.0\text{ mm}$  apart illuminate a screen  $1.5\text{ m}$  away. For light of wavelength  $500\text{ nm}$ , what is the fringe spacing on the screen?

- |                           |                           |
|---------------------------|---------------------------|
| <b>A</b> $0.25\text{ mm}$ | <b>B</b> $1.5\text{ mm}$  |
| <b>C</b> $2.5\text{ mm}$  | <b>D</b> $0.75\text{ mm}$ |
| <b>E</b> $0.50\text{ mm}$ |                           |

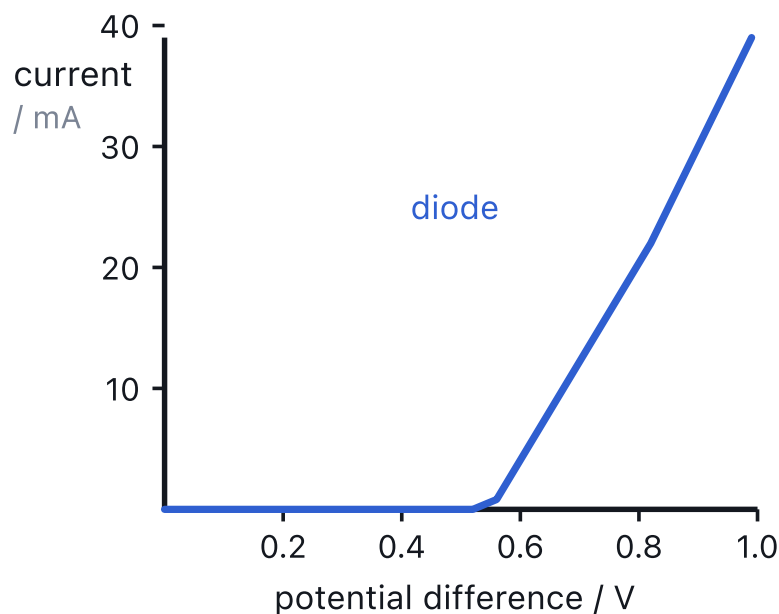
**Q19** · The critical angle: its sine for glass to air G

Light travels from glass of refractive index  $1.50$  to air ( $n = 1.00$ ). What is  $\sin$  of the critical angle for this interface?

- |                 |                 |
|-----------------|-----------------|
| <b>A</b> $0.67$ | <b>B</b> $1.50$ |
| <b>C</b> $0.50$ | <b>D</b> $0.33$ |
| <b>E</b> $1.00$ |                 |

**Q20** · A diode and a load resistor: the operating current from the load line H

A diode whose characteristic is shown is connected in series with a 100 ohm resistor and a 1.0 V supply.



What current flows? (On the drawn characteristic the current rises linearly from 0.55 V and passes through 22 mA at 0.8 V.)

- |                 |                 |
|-----------------|-----------------|
| <b>A</b> 10 mA  | <b>B</b> 22 mA  |
| <b>C</b> 4.0 mA | <b>D</b> 2.0 mA |
| <b>E</b> 6.0 mA |                 |

**Q21** · An RC charging circuit: the time constant I

A 2.0 microfarad capacitor is charged through a 1.0 megohm resistor by a 6.0 V supply. What is the time constant of the circuit?

- |                |                 |
|----------------|-----------------|
| <b>A</b> 12 s  | <b>B</b> 3.0 s  |
| <b>C</b> 2.0 s | <b>D</b> 0.50 s |
| <b>E</b> 1.0 s |                 |

**Q22** · Melting ice: the time from energy = power x time and latent heat J

A 1.0 kW heater is used to melt ice at 0 degrees C. How long does it take to melt 0.20 kg of ice? (Latent heat of fusion of ice =  $3.3 \times 10^5$  J/kg.)

- |                |                |
|----------------|----------------|
| <b>A</b> 33 s  | <b>B</b> 132 s |
| <b>C</b> 200 s | <b>D</b> 6.6 s |
| <b>E</b> 66 s  |                |

**Q23** · Mass-energy: energy released from a mass defect K

In a nuclear reaction 0.20 u of mass is converted into energy. Approximately how much energy is released? (Use  $1 \text{ u } c^2 = 930 \text{ MeV}$ .)

- A** 93 MeV
  - B** 186 MeV
  - C** 372 MeV
  - D** 46 MeV
  - E** 930 MeV
- 

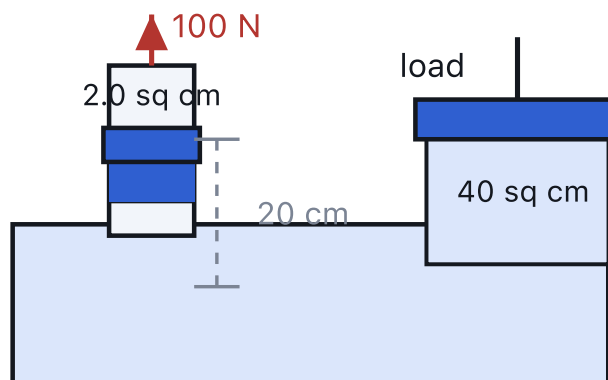
**Q24** · The photoelectric effect: maximum kinetic energy of the electrons L

Light whose photons each carry 5.0 eV of energy is incident on a metal whose work function is 2.0 eV. What is the maximum kinetic energy of the emitted electrons?

- A** 2.0 eV
  - B** 5.0 eV
  - C** 7.0 eV
  - D** 3.0 eV
  - E** 1.0 eV
- 

**Q25** · A hydraulic jack: the load lifted by the larger piston M

A hydraulic jack has a small piston of area  $2.0 \text{ cm}^2$  pushed down with a force of 100 N, and a large piston of area  $40 \text{ cm}^2$ .



What load can the large piston support?

- A** 2000 N
- B** 100 N
- C** 400 N
- D** 50 N
- E** 500 N